Zikai Zhou

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https://github.com/ZigZagKmp

EDUCATION

BS, Tsinghua University, Computer Science and Technology, 08/2022 - Present

- **Department:** Yao Class (Institute for Interdisciplinary Information Sciences)
- **GPA:** 3.9/4.0
- **Relevant Courses:** Cryptographic Protocols: Zero-Knowledge Proofs and MPC (A-), Fundamentals of Cryptography (A), Secure Multiparty Computation: Theory and Applications (A), Quantum Communication and Cryptography (A-), Mathematics for Computer Science (A), Abstract Algebra (A).

RESEARCH INTERESTS & SKILLS

Research Interests	Cryptography: FHE, ZKP, and MPC
Programming Languages	Familiar with C, C++, Python; Experience with Go, Matlab, Bash.

ACADEMIC EXPERIENCE

Student Research Intern @ Carnegie Mellon University 02/2025 - 07/2025

- Research Advisor: Prof. Wenting Zheng

Student Research Intern @ Tsinghua University 09/2025 - 01/2026

- Research Advisor: Prof. Yifan Song

Student Research Intern @ Tsinghua University 02/2026 - 05/2026

- Research Advisor: Prof. Mingyu Gao

Research Intern @ Ant Research 06/2026 - present

Teaching Assistant @ Tsinghua University 09/2024 - 01/2025

- Course: Cryptographic Protocols: Zero-Knowledge Proofs and MPC
- Class size: 20
- Course Instructor: Prof. Yifan Song
- Responsibilities: Graded and provided feedback on scribed lecture notes, graded assignments and the course project.

RESEARCH EXPERIENCE

Project: Optimal Bootstrapping Placement for Fully Homomorphic Encryption 10/2024 - 01/2026

- Led the design of a novel compiler for optimizing FHE circuits, focusing on optimal bootstrapping placement.
- Developed and implemented new algorithms to accelerate the search for optimal bootstrapping placements.
- Benchmarked the compiler against state-of-the-art algorithms and integrated FHE library backends for testing.
- **Co-first author** on a paper accepted by the *USENIX Security Symposium 2026*.

Project: Efficient honest majority MPC protocol with guaranteed output delivery 09/2025 - 01/2026

- Identified the core cost trade-off between authentication tag generation and authentication key generation.
- Exploring potential improvements.

Project: Efficient homomorphic RAM Framework via SIMD-FHE 02/2026 - 05/2026

- Formalized the variable-granularity fully homomorphic random access memory (FHE-RAM) framework.
- Designed and implemented an FHE-RAM utilizing a discrete CKKS scheme as the backend.
- Benchmarked the latency and throughput of the resulting FHE-RAM implementation.
- Investigating arithmetic and boolean scheme switching methodologies to further optimize FHE-RAM efficiency.

Project: Sparse-packed CKKS bootstrap implementation 04/2026

· Implemented sparse-packed CKKS bootstrap, achieving 1.03x - 2.35x acceleration.
· Contributed to Cipherflow's Lattisense repository.
· Won ecosystem contribution prize in Cipherflow Prometheus Program.

Project: Hardware Acceleration for FHE 06/2026 - present

· Understood MPU H1 hardware architecture.
· Investigating further optimization.

Course Project: Continuous LWE and its application in learning theory 04/2024 - 06/2024

Course: Fundamentals of Cryptography

· Reviewed the reduction from GapSVP and LWE to continuous LWE (group contribution).
· Investigated the reduction from certain learning problems to continuous LWE (CLWE).
· Generalized the cryptographic hardness of learning a cosine neuron to the hardness of learning any constant-period, constant-Lipschitz neuron.
· Unified the hardness proofs of different learning problems by summarizing the common crucial role of the homogeneous CLWE distribution (hCLWE).

RESEARCH PUBLICATION

Orbit: Optimizing Rescale and Bootstrap Placement with Integer Linear Programming Techniques
USENIX Security 2026 08/2026
Zikai Zhou*, *William Seo*, Edward Chen, Alex Ozdemir, Fraser Brown, Wenting Zheng*

AWARDS AND HONORS

Yao Award 2025 09/2025
Top 20% in Yao Class

National Scholarship 12/2023
Top 0.2% national-wide

Wang Dazhong Scholarship of Tsinghua 12/2025

Comprehensive Merit Scholarship of Tsinghua 12/2024
Sponsored by Ningbo Future Star Scholarship

Outstanding Student Leader of Tsinghua University 10/2024

Comprehensive Merit Scholarship of Tsinghua 12/2023
Sponsored by Institute for Interdisciplinary Information Core Technology Special Scholarship

Honorable Mention, Mathematical Contest in Modeling 02/2023

Gold Medalist, National Olympiad in Information Winter Camp (NOI-WC) 02/2021
In high school period

1st Prize, National High School Mathematics League 11/2021
In high school period

EXTRA-CURRICULAR

Senior Q&A Assistant 03/2023 - present
Tsinghua Drop-in Tutoring Center

· Assisted undergraduate students with problems in calculus, linear algebra, programming, and algorithms.

Voluntary Service: 500+ hours of volunteer service. Organized several institution-wide volunteer activities.